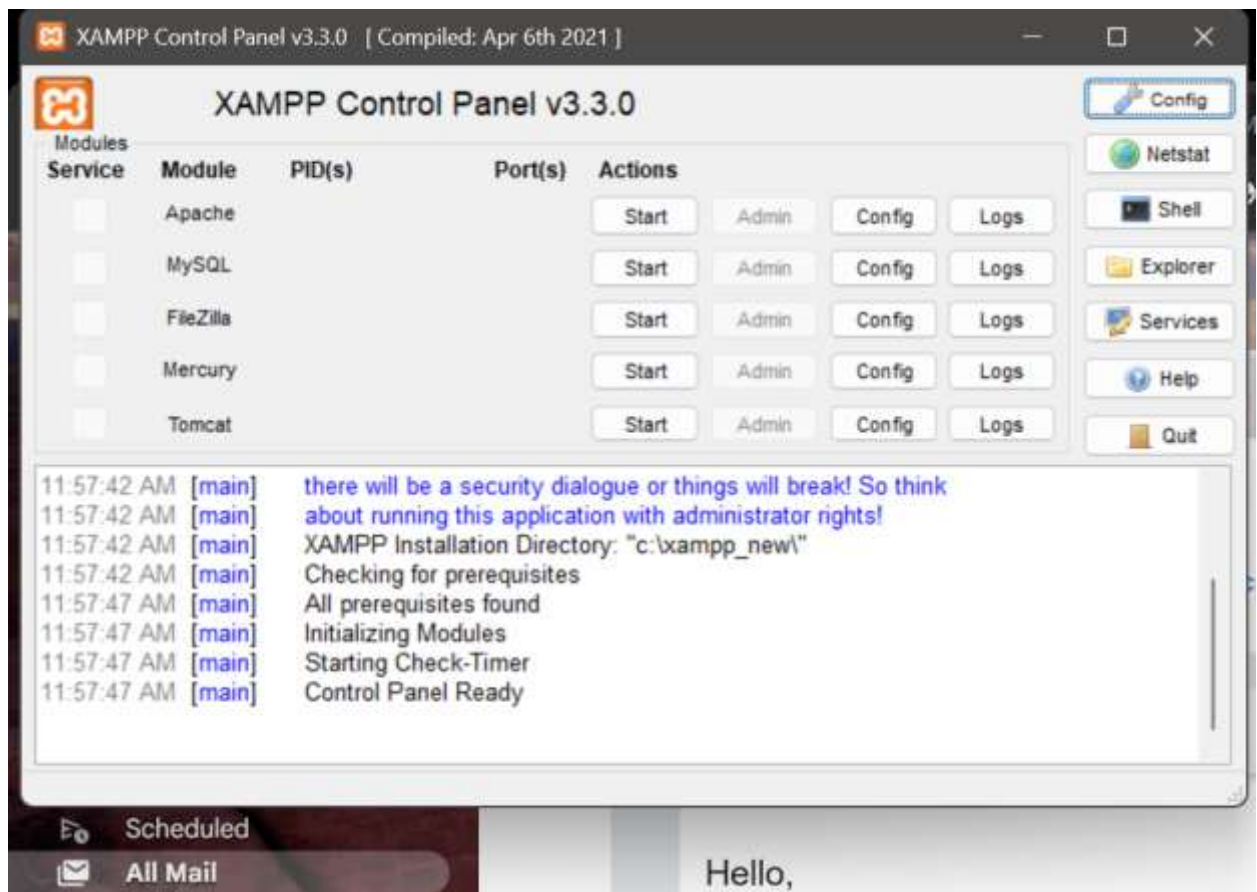


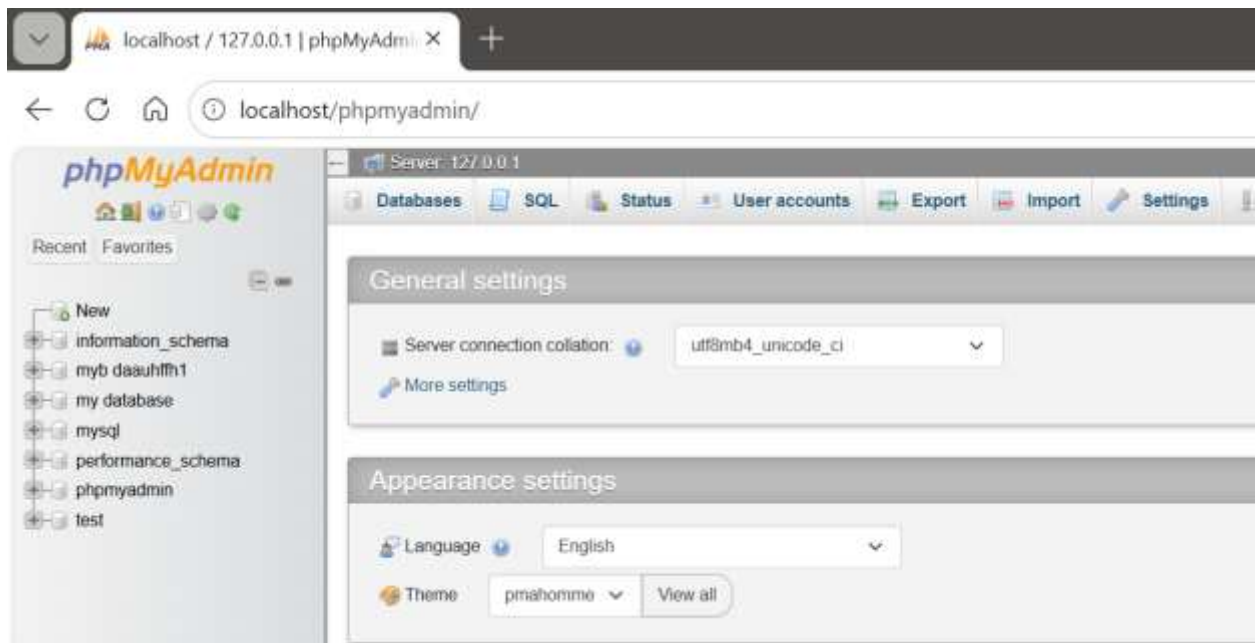
Session 1: Introduction to Databases
Trainer Coverage: What is a database, DBMS, RDBMS (with examples). Difference between MySQL, PostgreSQL, Oracle, SQLite. Tables, rows, columns, and relationships. Demo: Show data stored in MySQL Workbench.

Pending

Tasks

- 1) Install MySQL Workbench on your computer, connect to the local MySQL server, and take a screenshot of the default databases shown in the Schemas panel.





2) Create a new database called InstaClone in MySQL Workbench, then create a table named Users with columns: user_id (INT, primary key), username (VARCHAR), email (VARCHAR), and followers_count (INT).

Query:

Run SQL query/queries on database my databa:

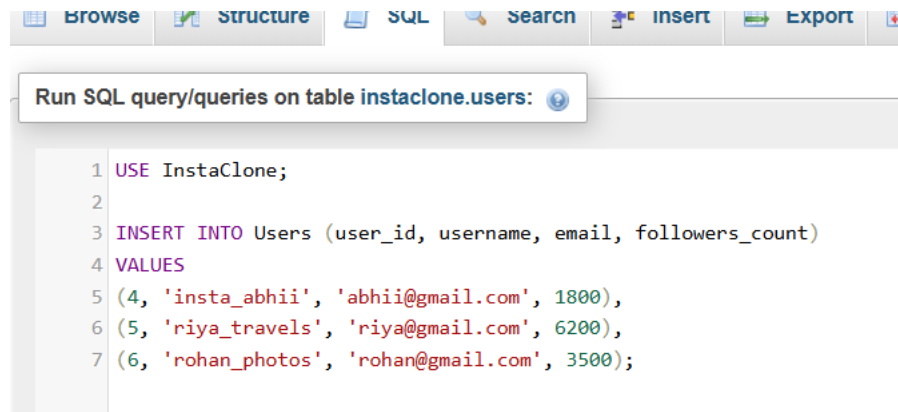
```
1 CREATE DATABASE InstaClone;
2
3 USE InstaClone;
4
5 CREATE TABLE Users (
6     user_id INT PRIMARY KEY,
7     username VARCHAR(100),
8     email VARCHAR(150),
9     followers_count INT
10 );
```

Output:

Table	Action	Index	Type	Collation	Size	Overhead
☐ users	☆ Browse					
1 table	Sum					
		#	InnoDB	utf8mb4_general_ci	16.0 K	-
		#	InnoDB	utf8mb4_general_ci	16.0 K	##

3) Insert 3 sample users into the Users table you created, using realistic Instagram-style usernames and follower counts.

Query:



The screenshot shows a SQL query editor with a toolbar at the top containing buttons for Browse, Structure, SQL, Search, Insert, and Export. Below the toolbar is a dropdown menu showing "Run SQL query/queries on table instaclone.users:". The query text is as follows:

```
1 USE InstaClone;
2
3 INSERT INTO Users (user_id, username, email, followers_count)
4 VALUES
5 (4, 'insta_abhii', 'abhii@gmail.com', 1800),
6 (5, 'riya_travels', 'riya@gmail.com', 6200),
7 (6, 'rohan_photos', 'rohan@gmail.com', 3500);
```

Output:

Extra options

				user_id	username	email	followers_count
☐	Edit	Copy	Delete	4	insta_abhii	abhii@gmail.com	1800
☐	Edit	Copy	Delete	5	riya_travels	riya@gmail.com	6200
☐	Edit	Copy	Delete	6	rohan_photos	rohan@gmail.com	3500

- 4) Create another table in the InstaClone database called Posts with columns: post_id (INT, primary key), user_id (INT), caption (VARCHAR), and post_date (DATE). Add a foreign key from Posts.user_id to Users.user_id to establish a relationship.

Query:

```
Run SQL query/queries on database instaclone: ⓘ

1  USE InstaClone;
2
3  CREATE TABLE Posts (
4      post_id INT PRIMARY KEY,
5      user_id INT,
6      caption VARCHAR(255),
7      post_date DATE,
8      FOREIGN KEY (user_id) REFERENCES Users(user_id)
9  );
```

Output:

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
post_id	int(11)	NO	PRI	NULL	
user_id	int(11)	YES	MUL	NULL	
caption	varchar(255)	YES		NULL	
post_date	date	YES		NULL	

5) Write a short comparison (3-4 lines each) of MySQL, PostgreSQL, Oracle, and SQLite, focusing on where each one is commonly used (for example: web apps, mobile apps, enterprise systems, etc.).

Database	Commonly Used For
MySQL	Widely used for web applications and websites. It is popular with PHP-based applications, content management systems, and e-commerce platforms. It is easy to use and has strong community support.
PostgreSQL	Commonly used for complex web applications, analytics, and data-intensive systems. It supports advanced SQL features and complex queries. It is popular in modern backend and enterprise applications.
Oracle Database	Mainly used in large enterprise systems, such as banking, healthcare, telecommunications, and government applications. It is designed for high performance, security, scalability, and large amounts of data.
SQLite	Commonly used in mobile apps, desktop applications, and embedded systems. It is lightweight and does not require a separate database server. It is especially popular for local storage in Android/iOS applications.